more general, namely

$$\alpha_u|u\rangle + \alpha_d|d\rangle.$$

If we include the apparatus as part of the system, the initial state is

$$\alpha_u|u, b\rangle + \alpha_d|d, b\rangle. \qquad (7.30)$$

This initial state is a product state, specifically a product of the initial spin state and the blank apparatus state. You can check that it is completely unentangled.

> **Exercise 7.10:** Verify that the state-vector in 7.30 represents a completely unentangled state.

Because we know from Eqs. 7.28 and 7.29 how the individual terms in 7.30 evolve, we can easily determine the final state:

$$\alpha_u|u, b\rangle + \alpha_d|d, b\rangle \rightarrow \alpha_u|u, +1\rangle + \alpha_d|d, -1\rangle.$$

This final state is an entangled state. In fact, if $\alpha_u = -\alpha_d$, it is the maximally entangled singlet state. Indeed, one can look at the apparatus and immediately tell what the spin state is: if the apparatus reads +1, the spin is *up*, and if it reads −1, the spin is *down*. Moreover, the probability that the final apparatus shows +1 is

$$\alpha_u^* \alpha_u.$$

This number represents a probability—it's exactly the same as the original probability that the spin was up. In this description of a measurement, no collapse of the wave function takes place. Instead, entanglement between the